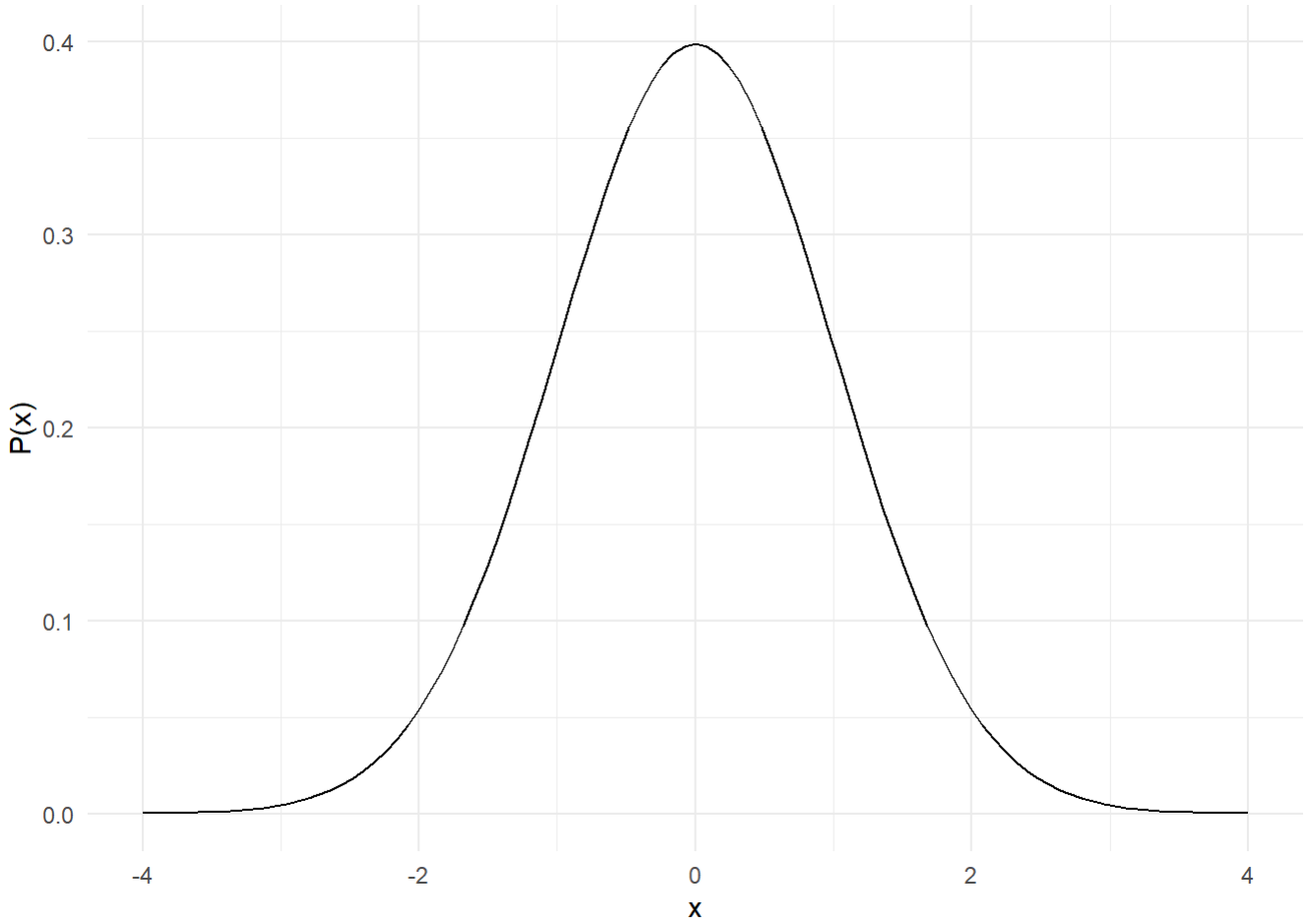
Lecture 6 Notes

Quantitative Data Analysis

Equations that changed the world

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

The Normal / Gaussian Distribution



Introduction

- We have been exploring wages and the wage gap
- We have observed a gap
- Our data covers only a sample of the population
- What if had a different sample?
- Presumably our new sample wouldn't be identical?
- Might the wage gap disappear if we had a different sample?

Sampling error

- This is a fundamental question in statistics
- We may have a finding in our sample
- However, if we have drawn a sample which isn't representative of the population then we will draw erroneous conclusions
- We refer to this a *sampling error*
- Let's have a look at our data

Mean wages for men and women

```
lfs %>% group_by(SEX) %>%
  summarise(mean_wage = mean(HOURPAY) %>% round(2)) %>%
  knitr::kable()
```

SEX	mean_wage
Male	22.48
Female	19.42

A different sample

- If we had a different sample, we might get different estimates of the mean
- We only have one sample (although we could triangulate with other data sources)
- As a thought experiment, let's suppose this data is the population
- Let's draw random samples from it and see what they might look like
- R will allow us to do this. Don't worry about the code, you only need to look at the results
- We will draw 5 samples each with 200 people

Parallel universes

Sample	Male	Female	Difference
1	22.79	19.69	-3.10
2	22.96	19.33	-3.63
3	24.89	19.17	-5.72
4	23.34	18.55	-4.79
5	23.51	18.09	-5.42

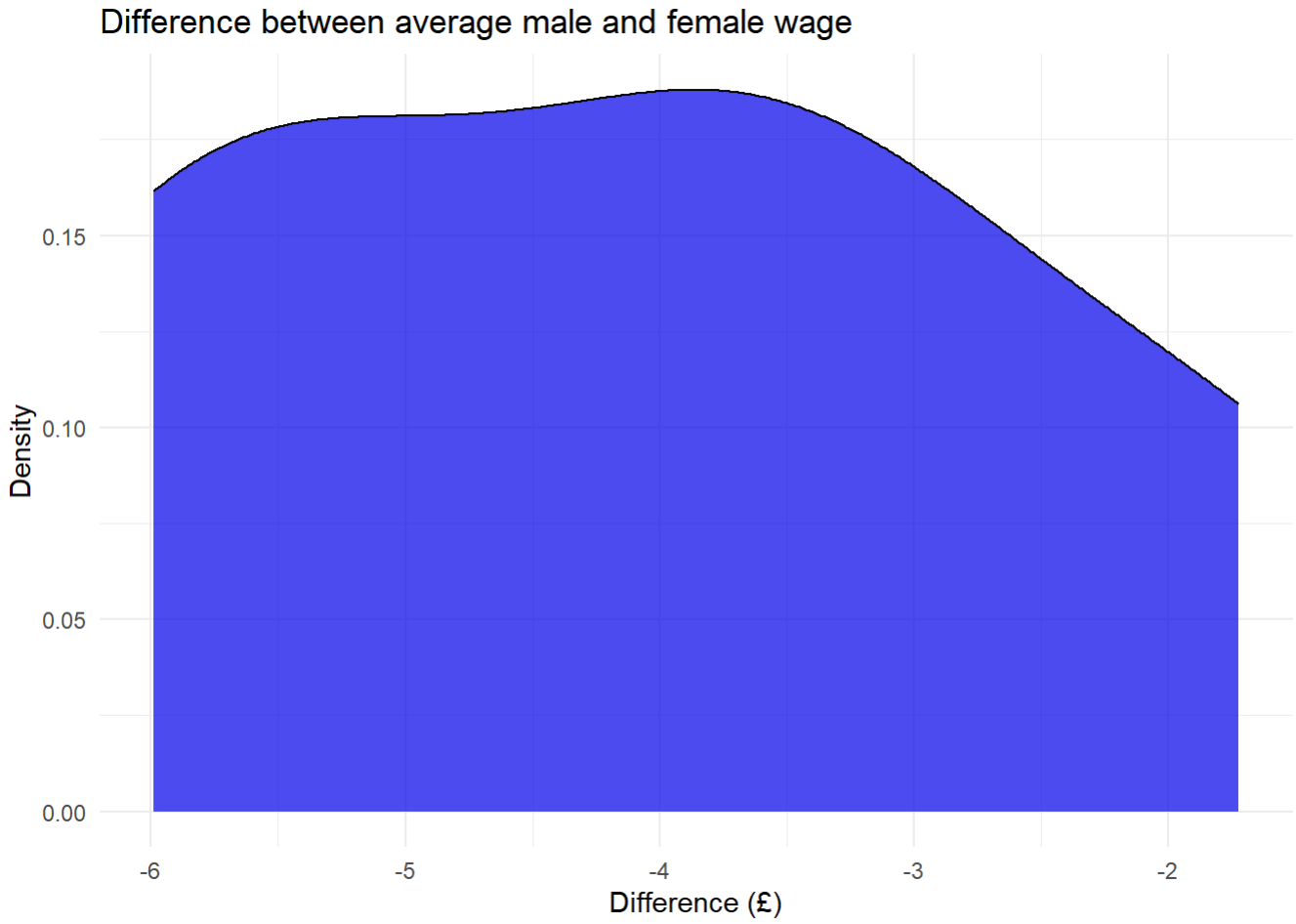
The problem

- In each of these parallel universes, a different sample is drawn
- We are in one of these parallel universes and have only observed one sample
- How do we know which one we are in?
- Are we in one with a representative sample?
- Are we in one which is horribly unrepresentative?

Plotting the parallel universes

- Let's begin by plotting the differences we observe from our different samples
- We'll start by plotting this for our 5 samples
- We'll then add more and more samples to it

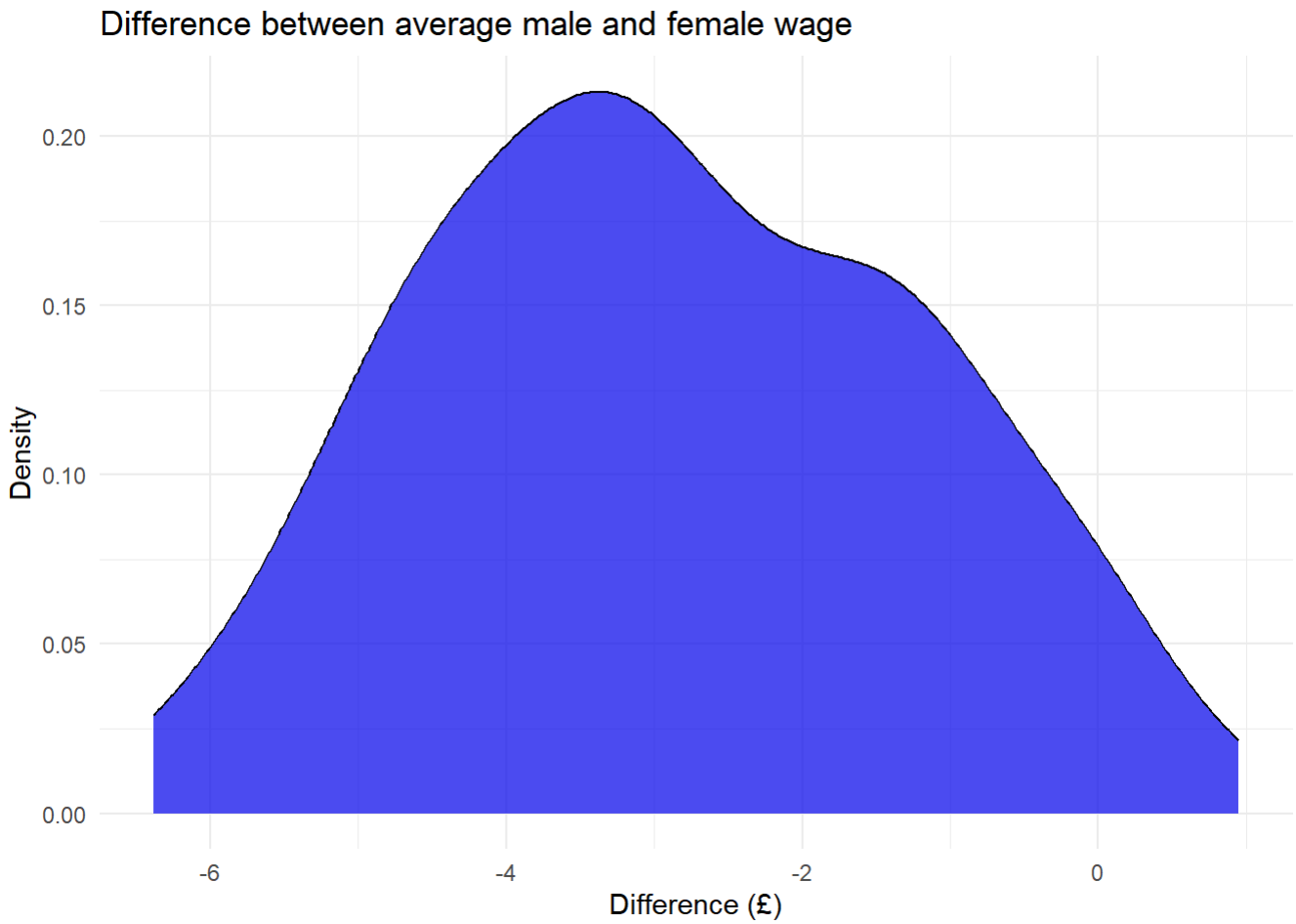
Results for 5 samples



Results for 50 samples



Resutls for 100 samples



Results for 1000 samples



Results for 1000 samples



Magic

- Hopefully you recognise this from the start of the lecture
- It turns out that when we calculated the difference based on repeated samples, the estimates follow a normal distribution
- Furthermore, is we calculate the mean of a variable based on a sample and then draw repeated samples, those estimates will be normally distributed
- This phenomenon is known as *central limit theorem* and is the basis of statistical inference

What can we do with this?

- The magic of central limit theorem allows us to say something about how certain we are
- In this case, we want to know whether we think the observed difference in male and female wages is due to sampling error
- Put another way, we want to know whether the true wage gap could be zero
- Without observing the population, we won't be able to say for sure
- However, we can make a statement about the *probability*
- Before that, let's say a few words about what probability is

Bayesian verus frequentist probability

- In this course (as in most courses like this) we will discuss probability from a classical or *frequentest* perspective
- We could also have conceptualised of probability in a a Bayesian way
- The frequentest approach conceptualises of probability as a frequency
- To see this in action, let's look at a frequency table of our sex variable

SEX	n	Proportion
Male	3410	0.46
Female	3937	0.54

Probability as a frequency

- I am going to randomly select a person from our population of interest
- What is the probability that they are male?
- We can read this from our frequency table by looking at what proportion of the people are male
- That is our best guess about the probability of randomly selecting a male from our population
- This is the key point of probability
- There are a set of rules governing the use of probabilities in calculations
- As we are doing an applied course, we don't need to concern ourselves too much with the details

Axioms and corollaries of probability

- $0 \leq P(A) \leq 1$
- $P(\Omega) = 1$
- $P(\emptyset) = 0$
- $P(\bar{A}) = 1 - P(A)$
- $P(A_1 \cup A_2) = P(A_1) + P(A_1)$, if A_1 and A_2 are disjoint
- $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$
- $P(A) \leq P(B)$, if $A \subseteq B$

Rule 1

- $0 \leq P(A) \leq 1$
- A is an event which may occur

- The probability of A occurring, P(A), is constrained to be between 0 and 1
- By definition, all probabilities must be between 0 and 1
- If you get a probability which is less than zero or greater than 1, something has gone wrong

Rule 2

- $P(\Omega) = 1$
- Ω is known as the sample space
- It is the set of *all* possible outcomes
- Subsets of the sample space are known as events
- We can therefore say that $P(\Omega) = 1$
- At least some event in the sample space will occur
- Certain events have a probability of 1

Rule 3

- $P(\emptyset) = 0$
- Ω is the set of all events
- $\emptyset$ is the empty or null set
- There are no events in the empty set, so the probability must be zero
- Impossible events have a probability of 0 i.e., they cannot and will not occur

Rule 4

- $P(\bar{A}) = 1 - P(A)$
- If event A occurs with probability $P(A)$
- Then A won't occur with a probability of $1 - P(A)$
- That is, the probability of *Not A* is given by $P(\bar{A}) = 1 - P(A)$
- $\bar{A}$ is known as a complementary event

Rule 5

- $P(A) \leq P(B)$, if $A \subseteq B$
- $\subseteq$ is the notation for subset
- This rule states that if A is a subset of B, then A cannot be more likely than B
- Suppose the probability that there will be rain tomorrow is 0.6

- The probability that there will be rain AND wind cannot be greater than 0.6

Rule 6

- $P(A_1 \cup A_2) = P(A_1) + P(A_2), A_1 \cap A_2 = \emptyset$
- This is one of the rules governing how we combine probabilities
- In set theory, $\cup$ represents a union and $\cap$ represents the intersection
- $A_1 \cup A_2$ are the objects belonging to set A_1 or A_2
- If the events cannot occur at the same time (i.e., they are mutually exclusive), we get the rule $P(A_1 \cup A_2) = P(A_1) + P(A_2)$

Rule 7

- $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$
- $A_1 \cap A_2$ is the probability that A_1 and A_2 happen at the same time
- If $A_1 \cap A_2 \neq \emptyset$ then is we want to know $P(A_1 \cup A_2)$, follow the rule $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$
- If we don't subtract $P(A_1 \cap A_2)$ then we double count

Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Bayes Theorem

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

What do we know?

- We know that repeated samples of the wage gap will be normally distributed
- Areas under a density curve can be interpreted as probabilities
- To make the calculations easier, we can standardise our estimate of the wage gap
- This will allow us to make use of the standard normal distribution
- The standard normal distribution has a mean of zero and a standard deviation of one
- If our observed wage gap is only due to sampling error, then the mean should be zero

- We therefore subtract zero from our estimate and divide by the standard deviation of the difference

The standard deviation of the difference

- If we know the population standard deviation for the groups we are studying, we can calculate the SE of the difference using:

$$se_{\bar{X}_M - \bar{X}_F} = \sqrt{\frac{\sigma_M^2}{N_M} + \frac{\sigma_F^2}{N_F}}$$

- σ_i^2 - Population variance of group i

- N_j - Sample size of group j

Drawing this together

- We will calculate a so-called Z score by
  - Subtracting the expected mean (assumed to be zero)
  - Dividing by the standard error
  - The resulting value will follow a standard normal distribution

$$Z = \frac{\bar{X}_M - \bar{X}_F}{se_{\bar{X}_M - \bar{X}_F}}$$

Problem

- We don't know the population standard deviation
- Let's gloss over this for the moment and use the sample standard deviations
- We will return to this point soon

SEX	n	mean_wage	var_wage
Male	3410	22.48	161.21
Female	3937	19.42	104.63

Calculating our Z value

$$Z = \frac{(22.48 - 19.42) - 0}{\sqrt{\frac{161.21}{3410} + \frac{104.63}{3937}}} = 11.26$$

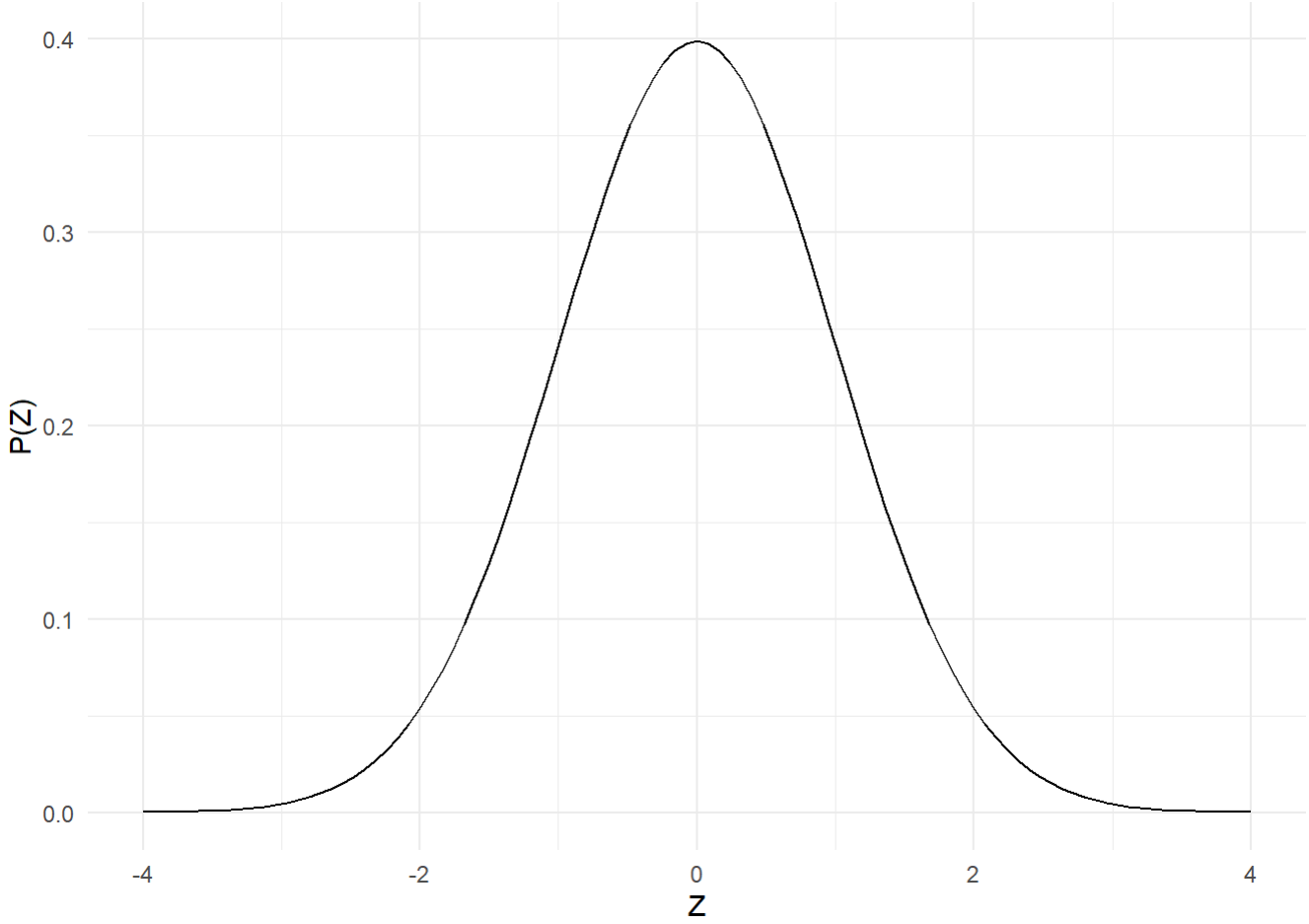
Interpretation

- If the true difference in wages is zero:

    - then the Z value should be close to zero
    - we are unlikely to observe anything far from zero
- If we observe a large difference, then we are in one of two worlds

1. There is no difference in wages, and we have drawn a very strange sample
2. There is a difference in the means, and our observed Z value is not improbable

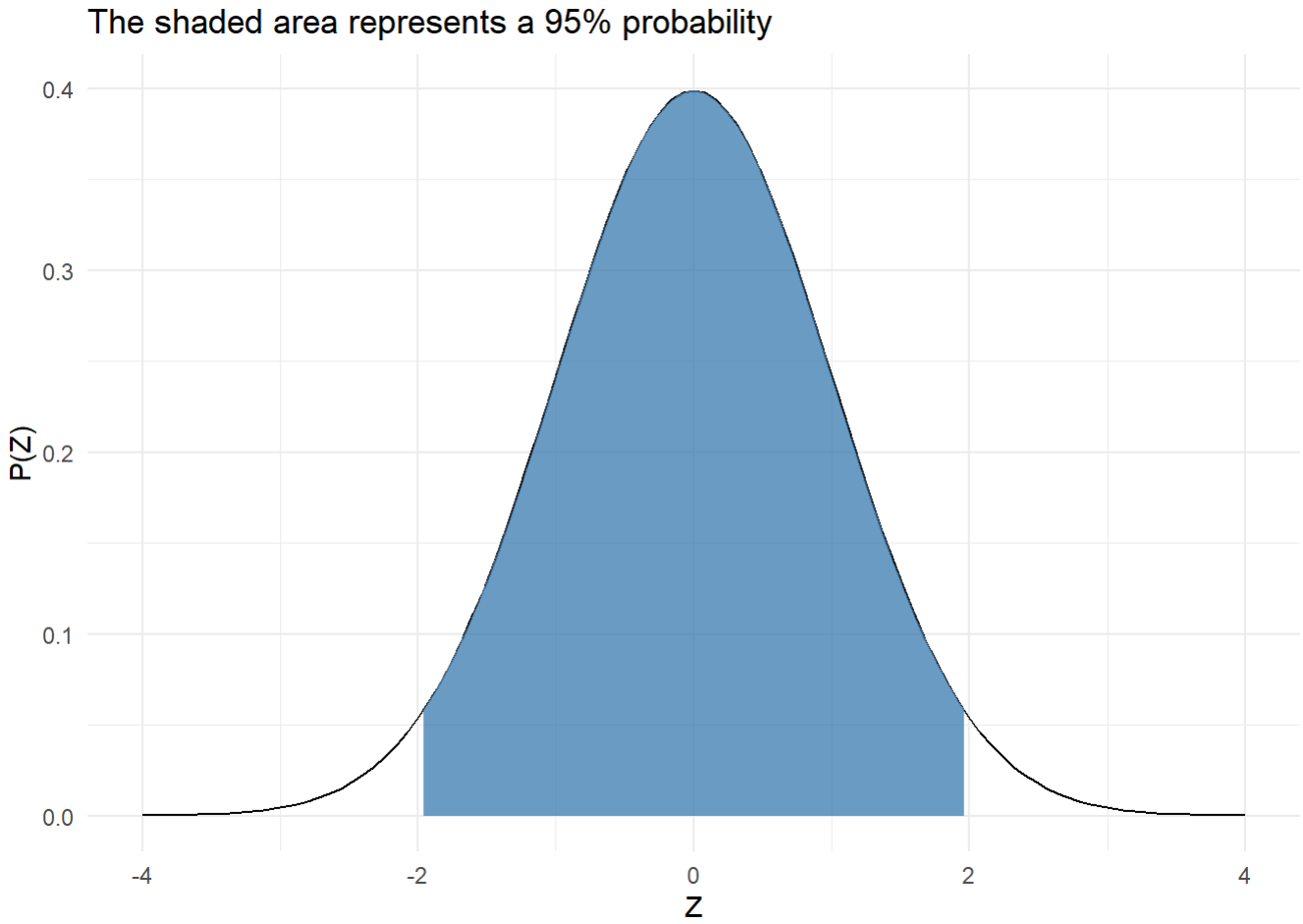
The standard normal distribution



What can we see?

- The bulk of the distribution is around zero
- The chance of observing anything less than minus two or greater than two looks pretty small
- In fact, if the wage gap in the population is zero, then there is a 95% chance that we will get a Z Value between -1.96 and 1.96
- In our case, the Z value is 11.26

The standard normal distribution



Calculating the probability

- We want to know whether the wage gap we observed is due to sampling error
- If the gap in the population is zero, then the curve we just saw lets us see the probabilities of observing different Z values
- We want to know the probability of observing a Z value as or more extreme than the one we observed
- We could do this with calculus, or ask R
- In the olden days we looked up the probabilities in books

Getting the probability from R

```
1-pnorm(11.26)
```

```
## [1] 0
```

- In our case, the probability is so low that R has rounded it to zero
- The probability isn't really zero (we could be wrong) but it is very small

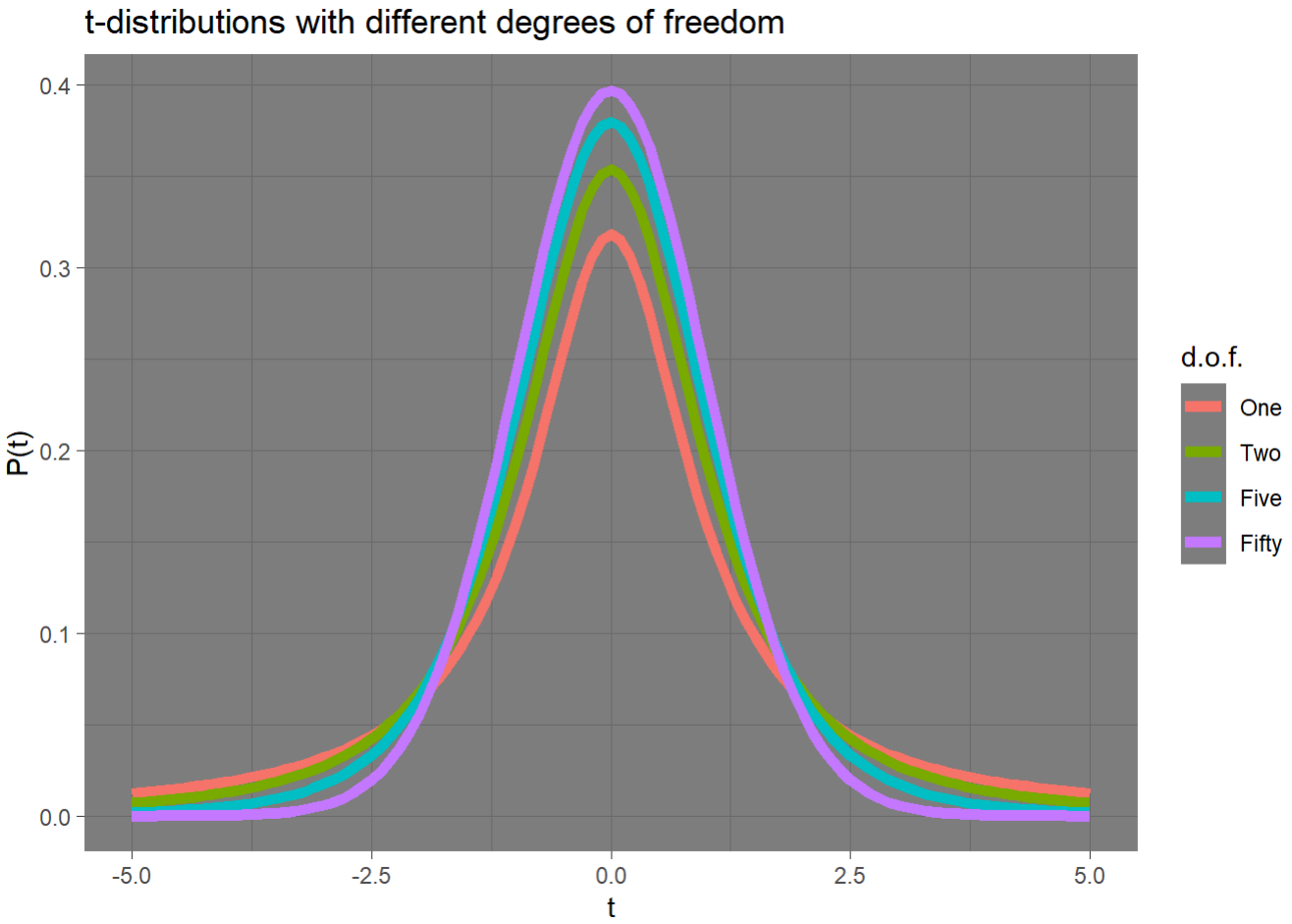
Making a decision

- In this case, there is almost no chance of observing a Z value of 11.26 if there is no wage gap in the population
- It is more reasonable to suppose that there is a difference in the population
- We will meet cases where the probability is not so low
- It is a convention that if the probability is less than or equal to 0.05, then we reject the hypothesis that the observed difference is due to sampling error
- We refer to 0.05 as the *level of significance* and sometimes use the Greek letter α to represent this concept

Population standard deviations

- Earlier we noted that this Z test assumed we knew the population standard deviations
- We don't know this so we used the sample standard deviation
- This introduces some additional uncertainty which skews our probability calculations
- A clever chap from the Guinness Brewery in Dublin figured out a solution
- Instead of using the standard normal distribution we use a t-distribution
- The shape of the t-distribution varies depending on the sample size (more precisely, the *degrees of freedom*)
- With a large enough number of observations, the t-distribution becomes the standard normal distribution

t-distribution



The t-test

- We can ask R to calculate the t-value and the associated probability
- This is known as the *t-test*
- Before conducting a test it is important that we are clear about what we are testing

    1. What is our null hypothesis
    2. What is our alternative hypothesis
    3. What is our level of significance

The null hypothesis

- Our hypothesis is that there is a difference between average wages of men and women in the population
- We test this by defining a null hypothesis which states that they are equal
- We then try to disprove this by showing such a belief is inconsistent with the data we have gathered
- We can write our null hypothesis as:

$$H_0 : \mu_{Male} = \mu_{Female}$$

The alternative hypothesis

- The alternative hypothesis is what we say we believe is true if we *reject* our null hypothesis
- There are three alternatives we may wish to consider:

1. $H_A : \mu_{Male} \neq \mu_{Female}$
2. $H_A : \mu_{Male} < \mu_{Female}$
3. $H_A : \mu_{Male} > \mu_{Female}$

- We will generally work with alternative hypotheses like the first one, a so-called two-sided or two-tailed test
- If we believe that the difference can only occur in one direction, then we may use the second or third

The level of significance

- There isn't really much choice to be made here
- It is a convention that we use a significance level of 0.05
- We will *reject* a null hypothesis if the probability, which we will call a *p-value*, is less than or equal to 0.05
- Otherwise we *fail to reject* the null hypothesis (we *never* accept a null hypothesis)

Conducting the t-test in R

- R has a special function for conducting the t-test, helpfully called *t.test()*
- We can specify the arguments in a number of ways
- I will suggest using formula notation: *HOURPAY ~ SEX*
- This will test how the mean of our dependent variable *HOURPAY* varies by our independent variable *SEX*
- We must also supply a *data* argument to show the function where to find the variables

Testing for differences in wage by sex

```
t.test(HOURPAY ~ SEX, data=lfs)
```

```
##
##  Welch Two Sample t-test
##
## data:  HOURPAY by SEX
## t = 11.257, df = 6531.3, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Male and group Fe
male is not equal to 0
## 95 percent confidence interval:
##  2.526349 3.591807
## sample estimates:
##   mean in group Male mean in group Female
##             22.48338             19.42430
```

Scientific notation

- Scientific notation is a way of expressing very small or very large numbers
- Any number can be written in scientific notation
- We take a number and multiply it by 10 raised to some power
- 1,530,000 can be written as 1.53×10^6
- 0.000000023 can be written as 2.3×10^{-8}
- The exponent tells us by how much, and in which direction, to move the decimal point
- You can turn it off in R by running *options(scipen = 999)*
- It can be enabled again by running *options(scipen = 0)*

Interpretation

- Our null hypothesis is that the population means are equal
- We calculated a t-value of 14.14
- If the null hypothesis is true, then there is a probability of 0.0000 of observing a p-value as or more extreme than 14.14
- As this probability is (much) less than 0.05 so we *reject* our null hypothesis
- We don't think the difference in sample means is only due to sampling variation

Confidence intervals

- Note that we have a confidence interval reported in our output
- This tells us a feasible range for the parameter of interest (the difference in wages in our case)

- When confidence intervals are calculated in this way, 95 times out of 100 they will contain the population parameter
- Some like to think of the interval as meaning we are 95% confident that the parameter is contained in the range
- This is not correct however, and is closer to the interpretation of a *credible interval* from Bayesian statistics

Errors

- We might have been wrong when we rejected the null hypothesis
- If we reject a null hypothesis which is actually true then we commit a *Type I* error
- If we fail to reject a false null hypothesis then we commit a *Type II* error
- Generally a Type I error is worse, because when we reject a null hypothesis we are making a claim to knowledge
- Some people like to think of this in the context of criminal justice:
    - A Type I error is analogous to finding an innocent person guilty
    - A Type II error would be letting a guilty person go free

The t-test is special

- The t-test is a special case of the linear model
- We are going to use the linear model a lot in the remainder of the course
- It offers us the flexibility to construct complex, sophisticated models
- It's pretty easy to ask for the t.test using the linear model
- We will use the *lm()* function
- We have to pass the results to the *summary()* function

Why the linear model?

- The linear model expresses what is going on using the equation of a straight line
- You might remember that the equation for a straight line is expressed as $y_i = \alpha + \beta \cdot x_i$
- α shows where the line *intercepts* the vertical axis
- β shows the *slope* of the line
- In our case, we use α to represent the wage of the *reference* category of sex
- x is a dummy variable, which we define as being equal to zero when the person is male, and 1 when they are female

- β therefore represents the difference between Female and Male

The t-test in the linear model

```
lm(HOURPAY ~ SEX, data=lfs) %>% summary()
```

```
##
## Call:
## lm(formula = HOURPAY ~ SEX, data = lfs)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -16.033  -7.639  -3.384   4.256  79.396
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  22.4834     0.1959  114.76   <2e-16 ***
## SEXFemale    -3.0591     0.2676  -11.43   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.44 on 7345 degrees of freedom
## Multiple R-squared:  0.01748,    Adjusted R-squared:  0.01734
## F-statistic: 130.6 on 1 and 7345 DF,  p-value: < 2.2e-16
```

Interpretation

- There are a lot of numbers here
- The *Estimate* column gives us the parameters we are interested in
- The intercept is the mean when our dummy variable is equal to zero
- The average male wage is therefore 22.48
- The difference between the average male and female wage is -3.06
- The average female wage is equal to the intercept plus the female coefficient i.e., 19.42`

Interpretation

- The *Std. Error* gives the standard error and is a measure of the precision of the estimate
- Dividing the estimate by the standard error gives us, as before, a t-value
- The *Pr(>|t|)* column gives the associated p-value
- Each line is testing the null hypothesis that the true value of the estimate is zero

- The first line is testing whether the average male wage is zero
- This is not a particularly useful test for us

Interpretation

- The next line tests whether the difference between men and women is zero
- As before, we reject the null hypothesis and conclude that there is a difference
- Note that the results aren't identical to what we had before (the degrees of freedom are different)
- We will return to this point later

Summary

- We have taken our first steps in inferential statistics
- Our key question has been whether a difference observed in our data is generalisable to the population
- The rules of probability and central limit theorem have allowed us to estimate the probability that observations from our data are only due to sampling error
- The rest of the course will be expanding on this basic idea